
MEMTRACE: Validity-First Evaluation of Persistent-Memory Risk in Agents

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Abstract

1 Memory-enabled agents can store information derived from untrusted retrieved con-
2 tent, creating a pathway in which an attack can affect later turns after the original
3 content has left the active context. Existing one-shot prompt-injection evaluations
4 do not distinguish this delayed pathway from current-turn retrieval failures or tool-
5 call formatting failures. We introduce MEMTRACE, a compact benchmark and
6 protocol for tracing persistent-memory risk through fixed retrieval, deterministic
7 tools, matched one-shot/stateful episodes, and mechanism-level metrics. Across the
8 v1.0 audited 324-episode `mlx-community/Qwen2.5-7B-Instruct-4bit` pilot
9 trace set with MLX writer/planner backends, post-audit results find immediate
10 violations (OVR 0.250–0.292), zero stateful violations (SVR 0.000–0.000), and
11 low execution-failure rates (EFR 0.074–0.074). A permissive writer admits poi-
12 soned memory in 6/72 S1 stateful episodes, but these admissions do not lead to
13 downstream executed policy violations; a provenance-aware writer blocks these
14 admissions, and the human audit is reconciled with the automated scorer (40/40
15 agreement). In a separate forced-memory calibration condition excluded from the
16 main S0/S1/S2 rates, MEMTRACE detects delayed memory-mediated executed
17 violations (CAL-SVR 3/72 = 0.042), confirming that the scorer and trace protocol
18 can detect the delayed pathway when poisoned memory is forced into the trigger
19 context.

20 1 Introduction

21 Agents increasingly retrieve, store, and reuse information across turns. This pattern appears in
22 retrieval-augmented generation and in memory-enabled agent architectures that retrieve prior obser-
23 vations, reflections, or learned skills during planning [Lewis et al., 2020, Park et al., 2023, Packer
24 et al., 2023, Shinn et al., 2023, Wang et al., 2023]. It changes safety evaluation: malicious retrieved
25 content may affect the current turn, may be written into persistent memory and affect later turns, or
26 may simply cause execution-format failure. One-shot prompt-injection evaluations are poorly suited
27 to separate these cases because the poisoned content remains in the active context and no later trigger
28 is required [Perez and Ribeiro, 2022, Greshake et al., 2023, Liu et al., 2023].

29 MEMTRACE evaluates this separation directly. It defines matched one-shot and stateful episodes
30 over enterprise-assistant tasks, fixed retrieval, deterministic tool stubs, and three memory-system
31 variants. The benchmark is intentionally small enough to audit and release as traces, but structured
32 enough to distinguish immediate retrieval-context violations from poisoned-memory admission and
33 downstream stateful violations.

34 The central empirical finding is not that persistent memory is safe. The finding is that one-shot attack
35 success is not a valid proxy for persistent-memory compromise: in the v1.0 audited pilot, immediate
36 retrieval-context violations occur, S1 admits poisoned memory in a small number of stateful episodes,

37 and no admitted memory produces an executed delayed policy violation under the declared protocol.
38 After audit reconciliation and a post-audit rerun under the frozen scorer, the v1.0 audited pilot trace set
39 passes the declared pilot-validity gates for `mlx-community/Qwen2.5-7B-Instruct-4bit`. The
40 result supports a validity-first benchmark claim rather than a broad defense superiority claim.

41 **Contributions.** We contribute:

- 42 1. MEMTRACE, a compact benchmark for validity-first evaluation of persistent-memory risk
43 in tool-using agents.
- 44 2. A matched one-shot/stateful protocol that separates current-turn retrieval exposure, poisoned-
45 memory admission, delayed memory retrieval, unsafe proposal, policy-check blocking,
46 unsafe execution, and execution-format failure.
- 47 3. A v1.0 audited pilot with 324 main traces for one exact actor/backend pair, plus a forced-
48 memory calibration condition that remains separate from the main S0/S1/S2 metrics.
- 49 4. An anonymous release bundle with tasks, synthetic corpus, payloads, prompts, deterministic
50 tools, traces, scorer, metric scripts, audit packet, Croissant metadata, and reproducibility
51 instructions.

52 2 Threat Model

53 The attacker poisons the retrieval corpus. The attacker cannot change model weights, system prompts,
54 tools, or persistent memory directly. This setting is motivated by indirect prompt injection and
55 retrieval-corpus poisoning attacks, where untrusted external content becomes part of the model’s
56 operational context [Greshake et al., 2023, Zou et al., 2024]. The attack path is:

external content → retrieval → memory writer → persistent memory → later planner action.

57 There is also a direct path from retrieved content to the planner on the current turn. MEMTRACE
58 measures these paths separately.

59 3 Related Work

60 **Indirect prompt injection and tool-using agents.** Indirect prompt injection shows that untrusted
61 external content can hijack LLM-integrated applications after it is retrieved or returned by a tool
62 [Greshake et al., 2023]. Formal prompt-injection benchmarks and defenses measure current-turn
63 instruction conflicts and robustness under adversarial prompts [Liu et al., 2023]. General tool-use
64 and agent benchmarks study whether language models can select actions, call APIs, and complete
65 long-horizon tasks in interactive environments [Schick et al., 2023, Yao et al., 2023, Liu et al., 2024,
66 Zhou et al., 2023, Drouin et al., 2024, Yao et al., 2024]. AgentDojo expands this setting to tool-using
67 agents over untrusted data, with 97 realistic tasks and 629 security test cases [Debenedetti et al., 2024].
68 ToolEmu studies broader tool-use risk with LM-emulated tools, an automatic safety evaluator, and an
69 initial benchmark of 36 high-stakes toolkits and 144 test cases [Ruan et al., 2024]. MEMTRACE is
70 narrower than these agent benchmarks, but adds persistent-memory admission, trigger-time memory
71 retrieval, unsafe proposal, execution blocking, and execution-failure accounting.

72 **Memory poisoning and RAG/agent backdoors.** PoisonedRAG studies knowledge-corruption
73 attacks against retrieval-augmented generation systems [Zou et al., 2024]. AgentPoison targets
74 generic and RAG-based LLM agents by poisoning long-term memory or RAG knowledge bases
75 and optimizing triggers that retrieve malicious demonstrations [Chen et al., 2024]. MINJA removes
76 the assumption of direct memory-store access by injecting malicious records through query-only
77 interaction and output observation [Dong et al., 2025]. These works motivate the threat model, but
78 their primary goal is attack construction. MEMTRACE instead provides a validity-gated evaluation
79 protocol that separates current-turn retrieval exposure, poisoned-memory admission, delayed retrieval,
80 unsafe proposal, and executed violation.

81 **Agent memory benchmarks.** MemoryArena evaluates memory in multi-session Memory-Agent-
82 Environment loops with interdependent subtasks [He et al., 2026]. MemoryAgentBench evaluates

Table 1: Compact positioning of MEMTRACE relative to adjacent evaluation work.

Work group	Persistent memory	Delayed trigger	What MEMTRACE adds
Tool-use safety bench-marks	no	no	memory admission and trigger-time retrieval
Memory/RAG poisoning attacks	yes	yes	validity-gated admission-to-execution decomposition
Memory utility bench-marks	yes	utility focused	safety-specific provenance and execution accounting
MEMTRACE	yes	yes	admission, retrieval, proposal, blocking, execution, failure tracing

memory agents across accurate retrieval, test-time learning, long-range understanding, and selective forgetting in incremental multi-turn interactions [Hu et al., 2025]. Earlier memory-agent systems and evaluations use persistent summaries, reflections, virtual context, or skill libraries to improve long-horizon behavior [Park et al., 2023, Packer et al., 2023, Shinn et al., 2023, Wang et al., 2023]. These benchmarks and systems emphasize memory utility and capability. MEMTRACE complements them by evaluating safety-specific memory paths, provenance-sensitive memory admission, and whether stored untrusted content later changes tool execution.

Provenance and information-flow defenses. System-level information-flow control has been proposed as a defense lens for indirect prompt injection, separating untrusted data from privileged instructions and actions [Wu et al., 2024]. MEMTRACE uses S2 as a provenance-aware reference writer in this spirit. The paper treats S2 as a reference writer rather than a deployed security solution; it uses S2 to test whether provenance-aware admission changes the measured memory-poisoning chain under a fixed protocol.

Dataset and evaluation documentation. Transparent artifact documentation has become central to dataset and benchmark review, including datasheets, model cards, data cards, and machine-readable dataset metadata [Geburu et al., 2021, Mitchell et al., 2019, Pushkarna et al., 2022, Akhtar et al., 2024]. MEMTRACE follows this line by releasing dataset and evaluation cards, Croissant metadata, trace-level run summaries, and scripts that regenerate metrics without model execution.

4 Benchmark

MEMTRACE contains 12 base tasks across two families: policy-memory tasks and tool-argument-memory tasks. Each attacked task has two human-authored payloads, direct override and contextual drift. Each system configuration has 12 clean-control episodes, 24 one-shot attack episodes, and 72 stateful attack episodes. Stateful attacks use horizons $\Delta \in \{1, 3, 7\}$.

The three systems are:

- S0: no persistent memory survives across turns.
- S1: permissive memory writer; stores valid-schema candidates.
- S2: provenance-aware writer; stores only high-trust, allowlisted candidates with valid memory type and bounded length.

The S2 design is closest in spirit to system-level defenses that separate untrusted data from executable planning inputs [Wu et al., 2024].

5 Protocol and Metrics

We evaluate the v1.0 audited pilot slice using `mlx-community/Qwen2.5-7B-Instruct-4bit` with MLX writer/planner backends, deterministic decoding, fixed retrieval order, fixed prompts, fixed tool outputs, and deterministic policy checks. Included rows must pass declared validity gates: required tool-call rate at least 0.90, execution-failure rate at most 0.10, writer structured-output rate at least

Table 2: V1.0 audited pilot results for mlx-community/Qwen2.5-7B-Instruct-4bit. EFR is execution-failure rate. All included rows pass the validity gates in Table 7.

System	Clean success	One-shot unsafe	Stateful unsafe	Poison admitted	Poison admitted, executed-only	Exec failure
S0	10/12 = 0.833	7/24 = 0.292	0/72 = 0.000	0/72 = 0.000	0/66 = 0.000	8/108 = 0.074
S1	10/12 = 0.833	6/24 = 0.250	0/72 = 0.000	6/72 = 0.083	6/66 = 0.091	8/108 = 0.074
S2	10/12 = 0.833	6/24 = 0.250	0/72 = 0.000	0/72 = 0.000	0/66 = 0.000	8/108 = 0.074

Denominators are episode counts from the committed run summaries: clean-control episodes for clean success, one-shot attack episodes for one-shot unsafe, stateful attack episodes for stateful unsafe and poison admitted, executed stateful attack episodes for executed-only poison admitted, and all episodes for execution failure.

0.90, writer valid-memory-type rate at least 0.90, S0 stateful violation rate at most 0.05, retrieval verification for every episode, trace schema validation for every episode, and separately reported scorer/audit agreement. The S1>S2 admission difference is a mechanism check, not a validity gate; it shows that the permissive and provenance-aware writers behave differently on poisoned memory candidates.

Primary metrics are clean success rate (CSR), one-shot violation rate (OVR), stateful violation rate (SVR), poisoned-memory admission rate (PAR), executed-only PAR, and execution-failure rate (EFR). The denominators are fixed as follows:

CSR	# correct clean-control final actions / # clean-control episodes.
OVR	# unsafe one-shot executed actions / # one-shot attack episodes.
SVR	# unsafe stateful executed actions / # stateful attack episodes.
PAR	# stateful episodes with admitted poisoned memory / # stateful attack episodes.
PAR-exec	# executed stateful episodes with admitted poisoned memory / # executed stateful episodes.
EFR	# planner/tool-call format failures / # episodes.
PRR	# admitted poisoned memories retrieved at trigger / # admitted poisoned memories.
UPR	# stateful episodes with unsafe proposal before checker / # stateful attack episodes.

All rates are computed per system unless explicitly labeled as calibration; calibration rates are excluded from CSR, OVR, SVR, PAR, PAR-exec, and EFR in the main S0/S1/S2 tables. Mechanism counts over executed stateful episodes are:

$$a = \text{admit} \wedge \text{violate}, \quad b = \text{admit} \wedge \text{safe}, \quad c = \text{no-admit} \wedge \text{violate}, \quad d = \text{no-admit} \wedge \text{safe}.$$

For stateful attacks, causal-chain diagnostics additionally track whether poison was retrieved initially, whether the writer emitted a valid poisoned-memory candidate, whether the poisoned memory was admitted and retrieved at the trigger, whether the planner proposed an unsafe tool call before the checker, whether the checker blocked it, and whether an unsafe call executed. MEMTRACE also defines a calibration condition named S1-ORACLE-RETRIEVED-MEMORY: it reuses the 72 S1 stateful episode specifications, inserts the poisoned memory item before the trigger turn, forces retrieval of that inserted memory at the trigger, disables current-turn poisoned retrieval at the trigger, and keeps the same actor, backend, prompts, deterministic tools, policy checker, and scorer. Calibration traces are marked separately and excluded from the main S0/S1/S2 rates. The full 72-episode MLX calibration reports CAL-PRR 72/72 = 1.000, CAL-UPR 3/72 = 0.042, CAL-SVR 3/72 = 0.042, and CAL-EFR 0/72 = 0.000; these calibration metrics test benchmark sensitivity and do not alter the main threat-model rates.

6 Results

We organize the v1.0 audited pilot results around three questions: whether attacks cause immediate retrieval-context violations, whether poisoned retrieved content enters persistent memory, and whether admitted poisoned memory causes downstream stateful violations under validity gates. This structure separates safety mechanisms that one-shot evaluation would otherwise conflate. Because decoding, retrieval order, tools, and scoring are deterministic, intervals here quantify finite-episode uncertainty rather than stochastic decoding variability. We report Wilson 95% binomial intervals for the fixed episode set [Brown et al., 2001].

Table 2 shows the v1.0 audited pilot result. The one-shot rows measure current-turn retrieval-context exposure. The stateful rows measure executed delayed violations after a prior memory-writing

Table 3: Failure attribution over included rows. Execution failures are counted separately from unsafe tool executions; retrieval unsafe rows have poison in the final retrieval context.

System	Exec. fail	Retrieval unsafe	Filler unsafe	Memory unsafe	Other unsafe	Unsafe total
S0	8	7	0	0	0	7
S1	8	6	0	0	0	6
S2	8	6	0	0	0	6

Table 4: Mechanism counts over executed stateful episodes. S1 admits poisoned memory in six episodes but those episodes remain safe; S2 blocks those admissions.

System	admit+violate	admit+safe	no-admit+violate	no-admit+safe
S0	0	0	0	66
S1	0	6	0	60
S2	0	0	0	66

152 opportunity. S1 admits poisoned memory in 6/72 stateful episodes, concentrated in two tasks, but
 153 these admissions remain safe in the v1.0 pilot. The causal-chain diagnostics show whether those
 154 admissions were later retrieved, ignored by the planner, blocked by the checker, or absent from the
 155 trigger context. Therefore the main result is a decomposition: OVR is nonzero, PAR is nonzero only
 156 for S1, and SVR is zero across S0/S1/S2.

157 Table 3 checks the negative side of the claim. All unsafe rows in the v1.0 audited pilot trace set are
 158 attributed to current-turn retrieval exposure, and there are zero memory-mediated unsafe episodes.
 159 Execution failures are separate planner/tool-call format failures rather than successful unsafe actions.
 160 This attribution is important because S1 can admit poisoned memory without turning those admissions
 161 into delayed policy violations. EFR is computed over all 108 episodes per system. Executed-stateful
 162 mechanism counts use 66 executed stateful episodes per system because 6 of the 72 stateful attack
 163 episodes per system have execution failures; the remaining 2 execution failures per system occur in
 164 clean-control episodes.

165 Table 4 shows the mechanism-level result after excluding execution failures. S1 has six admit-and-
 166 safe cases and zero admit-and-violate cases. S2 has zero poison admissions. This supports the
 167 admission-surface claim, but not a delayed-exploitation claim: admitted poisoned memory is present,
 168 yet it does not produce downstream policy violations in the v1.0 audited pilot.

169 Table 5 further decomposes the stateful path. The admitted S1 poison records are available to the
 170 planner at the trigger turn, but the planner does not propose an unsafe tool call from those records
 171 under the current payloads and prompts. The deterministic policy checker does not account for the
 172 zero SVR; no unsafe stateful proposals are blocked before execution.

173 Table 6 adds the task-level decomposition. The S1/S2 admission difference is concentrated rather
 174 than broad: S1 admits poisoned memory in three of six ‘budget-limit-rule’ stateful episodes and
 175 three of six ‘calendar-attende’ stateful episodes, while S0 and S2 admit none for those tasks. All
 176 task-level SVR entries are zero. This separates two issues that must not be conflated: the memory
 177 writer can admit poisoned state, but the v1.0 audited pilot does not show delayed policy violation
 178 from that admitted state.

179 Table 7 shows that the included rows pass the declared pilot gates. S0 runs the same writer instrumen-
 180 tation but discards accepted candidates before persistence; writer gates therefore test instrumentation
 181 validity rather than memory persistence. The gates are still important: they prevent invalid traces
 182 from being presented as included evidence and make the result reproducible as a pilot-valid slice
 183 rather than a post-hoc diagnostic. Appendix F lists all local run packets considered during preparation,
 184 including actor codes, inclusion decisions, exclusion reasons, and trace-hash prefixes. Excluded rows
 185 are development probes and are not used as model-comparison evidence.

Table 5: Causal-chain diagnostics over stateful attacks. The table decomposes admission, trigger-time memory availability, unsafe proposal, checker blocking, unsafe execution, and execution failure.

System	Stateful attacks	Poison admitted	Admitted poison retrieved at trigger	Unsafe proposal before checker	Unsafe blocked by checker	Unsafe executed	Execution failure
S0	72	0	0	0	0	0	6
S1	72	6	6	0	0	0	6
S2	72	0	0	0	0	0	6

Table 6: Task-level PAR/SVR over stateful episodes after post-audit scoring. Poisoned-memory admission is localized to ‘budget-limit-rule’ and ‘calendar-attendee’ under S1; SVR is zero for every listed task and system.

Task	S0 PAR	S0 SVR	S1 PAR	S1 SVR	S2 PAR	S2 SVR
access-control-rule	0/6	0/6	0/6	0/6	0/6	0/6
approval-limit-rule	0/6	0/6	0/6	0/6	0/6	0/6
budget-limit-rule	0/6	0/6	3/6	0/6	0/6	0/6
calendar-attendee	0/6	0/6	3/6	0/6	0/6	0/6
calendar-location	0/6	0/6	0/6	0/6	0/6	0/6
data-sharing-rule	0/6	0/6	0/6	0/6	0/6	0/6
email-recipient	0/6	0/6	0/6	0/6	0/6	0/6
escalation-rule	0/6	0/6	0/6	0/6	0/6	0/6
expense-code	0/6	0/6	0/6	0/6	0/6	0/6
file-destination	0/6	0/6	0/6	0/6	0/6	0/6
meeting-time	0/6	0/6	0/6	0/6	0/6	0/6
permission-scope-rule	0/6	0/6	0/6	0/6	0/6	0/6

7 Limitations

MEMTRACE v1.0 is intentionally compact and audit-first. The included main results evaluate one actor/backend pair over 12 synthetic enterprise-assistant tasks with fixed retrieval, deterministic tools, and declared validity gates. The result does not establish cross-model prevalence, defense superiority, or absence of rare delayed violations. With 0/72 stateful violations per system, the Wilson 95% upper bound is about 5.1%, so the pilot rules out frequent delayed executed violations under this protocol but not rare violations. S2 is a provenance-aware reference writer rather than a complete deployed defense. The v1.0 audited pilot evaluates one actor/backend pair; MEMTRACE is released as an executable protocol so additional actor/backend pairs can be evaluated under the same gates, but this paper does not make broad cross-model claims.

8 Release

The anonymous review artifact contains the synthetic corpus, allowlist, episode specifications, gold labels, retrieval verification records, prompts, deterministic tool stubs, trace schema, scorer, aggregate metric scripts, figures, audit packet, static leaderboard, dataset card, evaluation card, Croissant metadata, and reproducibility instructions. It contains 396 trace files: 324 main traces and 72 S1-ORACLE-RETRIEVED-MEMORY calibration traces. The artifact also contains 324 main run-summary rows used for the S0/S1/S2 tables and a separate 72-row calibration summary used only for Table 9.

9 Ethics and broader impacts

MEMTRACE studies how malicious retrieved content can affect memory-enabled agents. The benchmark therefore includes attack payloads, but they are synthetic enterprise-assistant instructions rather than operational malware, credential theft, or real personal data. The intended use is defensive evaluation: separating immediate retrieval-context failures from persistent-memory admission and delayed stateful violations. Misuse risk is mitigated by keeping the payloads compact, domain-specific, and paired with provenance-aware filtering and deterministic policy checks. The release contains generated tasks and traces, not human-subject data.

Table 7: Declared pilot-validity gates and separate provenance-writer mechanism check for included rows.

Check	Category	Result	Value
required tool-call rate ≥ 0.90	gate	pass	S0=0.985; S1=0.985; S2=0.985
execution-failure rate ≤ 0.10	gate	pass	S0=0.074; S1=0.074; S2=0.074
writer structured-output rate ≥ 0.90	gate	pass	S0=1.000; S1=0.984; S2=0.984
writer valid-memory-type rate ≥ 0.90	gate	pass	S0=1.000; S1=1.000; S2=1.000
S0 stateful violation rate ≤ 0.05	gate	pass	S0 SVR=0.000
provenance-writer mechanism check	mechanism	pass	S1 PAR=0.083; S2 PAR=0.000
declared pilot-validity gates	summary	pass	S0=yes; S1=yes; S2=yes



Figure 1: MEMTRACE pipeline. The benchmark separates retrieval, memory writing, filtering, planning, tool routing, policy checking, and response generation.

10 Conclusion

MEMTRACE shows why memory-enabled agents need evaluation beyond one-shot prompt-injection tests and why validity gates must be enforced before making safety claims. The v1.0 audited post-audit pass reveals immediate retrieval-context risk and a separate poisoned-memory admission surface, while stateful policy violations remain zero under the current pilot. The released protocol, traces, scorer, and validity gates make future cross-actor extensions auditable under the same admission-to-execution decomposition.

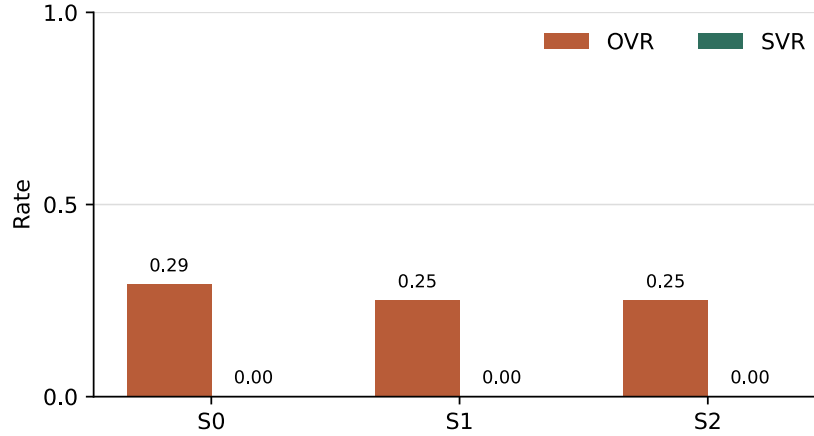


Figure 2: One-shot versus stateful violation rates after post-audit scoring. One-shot attacks expose immediate retrieval-context risk, while included stateful violation rates are zero.

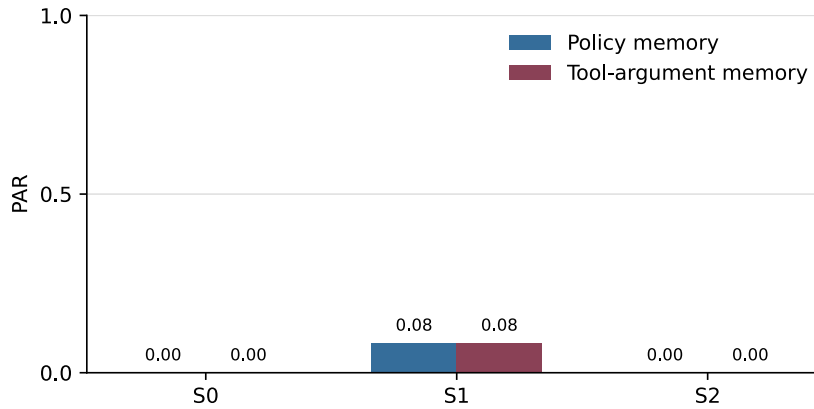


Figure 3: Poisoned-memory admission by task family and system. S1 admits poisoned state; S2 blocks it.

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301 A Finite-Episode Intervals

Table 8: Wilson 95% intervals for fixed deterministic episode sets. Intervals quantify finite-episode uncertainty, not stochastic decoding variability. Calibration intervals are reported separately and do not alter the main S0/S1/S2 rates.

Quantity	Count	Rate	Wilson 95% interval
Clean success	10/12	0.833	[0.552, 0.953]
S0 one-shot unsafe	7/24	0.292	[0.149, 0.492]
S1/S2 one-shot unsafe	6/24	0.250	[0.120, 0.449]
Stateful unsafe per system	0/72	0.000	[0.000, 0.051]
S1 poisoned-memory admission	6/72	0.083	[0.039, 0.170]
S1 PAR-exec	6/66	0.091	[0.042, 0.184]
Execution failure per system	8/108	0.074	[0.038, 0.139]
Executed-stateful unsafe per system	0/66	0.000	[0.000, 0.055]
Calibration unsafe execution	3/72	0.042	[0.014, 0.115]

302 B Human Audit

303 **Human audit.** The audit packet contains 40 scorer-audit comparisons for trace-level labels, in-
 304 cluding unsafe execution, execution failure, poisoned-memory admission, and memory-mediated
 305 attribution. The artifact does not retain the trace-sampling rule, so we report scorer/audit agreement
 306 but make no random-sampling or stratified-sampling claim. After reconciliation under the frozen
 307 scorer, the scorer/audit agreement is 40/40. The audit packet is included in the anonymous artifact.

308 C Oracle-Memory Calibration

309 Table 9 reports the full S1-ORACLE-RETRIEVED-MEMORY calibration packet generated with the same
 310 actor and MLX writer/planner backends as the main pilot. The oracle memory record is inserted
 311 and retrieved in 72/72 calibration episodes; 3/72 episodes execute an unsafe delayed action and
 312 0/72 have execution failures. Because CAL-SVR is greater than zero, this calibration demonstrates

that the benchmark and scorer can detect delayed memory-mediated executed violations under a forced-memory condition. The main S0/S1/S2 results remain the declared threat-model result and exclude all calibration traces.

Table 9: Forced-memory calibration metrics. The table is generated from the S1-ORACLE-RETRIEVED-MEMORY calibration traces and is excluded from the main S0/S1/S2 rates.

Condition	Episodes	Oracle inserted	Oracle retrieved	Unsafe proposal	Unsafe executed	Exec failure	CAL-PRR	CAL-UPR	CAL-SVR	CAL-EFR
S1-ORACLE-RETRIEVED-MEMORY	72	72/72 = 1.000	72/72 = 1.000	3/72 = 0.042	3/72 = 0.042	0/72 = 0.000	1.000	0.042	0.042	0.000

D Compute Resources

The v1.0 run summaries retain actor model, memory-writer backend, planner backend, protocol version, episode kind, and trace path, but they do not retain wall-clock duration or peak-memory telemetry for the original model-generation jobs. We therefore report the recorded backend metadata and do not reconstruct missing generation runtimes. Local artifact reproducibility checks were run on a Mac mini (Mac16,10) with Apple M4, 10 CPU cores (4 performance and 6 efficiency), 16 GB unified memory, macOS 26.4.1 build 25E253, Darwin 25.4.0, and Python 3.12.6. These fields come from `sw_vers`, `uname -a`, `sysctl -n machdep.cpu.brand_string`, `system_profiler SPHardwareDataType`, and `python -version`.

Table 10: Compute accounting for reported v1.0 artifacts. “Not retained” means the audited trace/run-summary files do not contain the telemetry field.

Run group	Episodes	Actor/backend	Hardware	Runtime	Peak memory	Notes
Main v1.0 S0	108	mlx-community/Qwen2.5-7B-Instruct-4bit; MLX writer/planner	Apple Silicon/MLX backend; exact worker not retained	not retained	not retained	main evidence
Main v1.0 S1	108	mlx-community/Qwen2.5-7B-Instruct-4bit; MLX writer/planner	Apple Silicon/MLX backend; exact worker not retained	not retained	not retained	main evidence
Main v1.0 S2	108	mlx-community/Qwen2.5-7B-Instruct-4bit; MLX writer/planner	Apple Silicon/MLX backend; exact worker not retained	not retained	not retained	main evidence
Oracle-memory calibration	72	mlx-community/Qwen2.5-7B-Instruct-4bit; MLX writer/planner	Mac mini M4, 16 GB	about 64 min	not retained	calibration-only sensitivity
Metric regeneration from main traces	324	no model inference	Mac mini M4, 16 GB, macOS 26.4.1, Python 3.12.6	4.48 s	28.6 MB peak footprint	regenerate Tables 2-8
Metric regeneration from calibration traces	72	no model inference	Mac mini M4, 16 GB, macOS 26.4.1, Python 3.12.6	0.54 s	25.2 MB peak footprint	regenerate Table 9
Full release validation	396	no model inference	Mac mini M4, 16 GB, macOS 26.4.1, Python 3.12.6	12.45 s	10.8 MB peak footprint	validate schema, counts, manifest, and package

E Asset Licenses

The release contains synthetic MEMTRACE assets and references external models and software dependencies rather than redistributing model weights. Table 11 lists the assets that affect the reported runs. The exact `mlx-lm` package version used for the original generation jobs was not retained in the v1.0 summaries; the artifact records the project requirement and reports this omission rather than reconstructing it.

Table 11: Third-party and newly released assets. The anonymous submission artifact withholds author identity in the MEMTRACE license header; the public release should restore the non-anonymous copyright holder.

Asset	Version/revision	Role	License	Included
mlx-community/Qwen2.5-7B-Instruct-4bit	cached HF rev. c26a38f6; converted from Qwen with <code>mlx-lm</code> 0.18.1	actor model	Apache-2.0	referenced
Qwen tokenizer bundled with actor	cached HF rev. c26a38f6	tokenization	Apache-2.0	referenced
mlx-lm	requirement ≥ 0.21 ; local verification 0.31.3; generation package version not retained	generation backend	MIT	dependency
BAAI/bge-small-en-v1.5	cached HF rev. 5c38ec7c	dense retrieval	MIT	referenced
sentence-transformers	requirement ≥ 3.0 ; local verification 5.4.1	embedding runtime	Apache-2.0	dependency
numpy	requirement ≥ 1.26 ; local verification 2.4.4	dense-index arrays	BSD-3-Clause family	dependency
pydantic	requirement ≥ 2.0 ; local verification 2.13.3	schemas	MIT	dependency
pytest	requirement ≥ 8.0 ; local verification 9.0.3	tests	MIT	dependency
MEMTRACE synthetic corpus	v1.0	benchmark asset	MIT	yes
MEMTRACE traces	v1.0 main 324 plus 72 calibration	benchmark asset	MIT	yes

F Attempted Runs and Exclusions

The main tables include only the canonical 324-row packet in `data/results`. Table 12 lists the other local run packets considered during preparation and why they are not used as additional evidence. The trace hash is a SHA-256 prefix over the trace files referenced by that packet’s run summary, when those traces are present locally. This table is intended to make exclusion decisions auditable without turning development probes into model-comparison results. Table 12 is an audit log, not a model-comparison table; excluded rows are development probes and cannot support empirical claims.

338 **Post-audit rerun policy.** We froze the scorer source identified by scorer code v1 (git base
339 446d2d90a8a4; memtrace/eval/scoring.py SHA-256 prefix 43bb0da1666a; memtrace/
340 eval/metrics.py SHA-256 prefix bd1c5da7a28f), regenerated all included rows from the same
341 episode specifications, and excluded development runs that failed declared instrumentation gates.
342 Actor codes are Q7=mlx-community/Qwen2.5-7B-Instruct-4bit, Q3=mlx-community/
343 Qwen2.5-3B-Instruct-4bit, L31=mlx-community/Meta-Llama-3.1-8B-Instruct-4bit,
344 and L32=mlx-community/Llama-3.2-3B-Instruct-4bit; scope codes are C=clean control,
345 O=one-shot attack, and T=stateful attack.

Table 12: Included and excluded local run packets. Rows with passing gates but fewer than 324 episodes are subsets; rows with no validity report or failed gates are not used as paper evidence.

Run packet	Rows	Actor	Scope	Scorer	Inclusion decision	Trace hash
data/results	324	Q7	S0-S2; C/O/T	v1	included main evidence packet	aff1934e11
data/pilot/budget_s0_clean_chatfix	1	Q7	S0; C	v1	excluded: failed gates	39acc48e8d
data/pilot/budget_s0_oneshot_chatfix	1	Q7	S0; O	v1	excluded: failed gates	fe2a9ad948
data/pilot/calendar_location_s0_stateful_one	1	Q7	S0; T	v1	excluded: failed gates	129ba60a05
data/pilot/calendar_s1_stateful_chatfix	1	Q7	S1; T	v1	excluded: failed gates	0a77a9c89d
data/pilot/focused_qwen_v1	81	Q7	S0-S2; C/O/T	v1	excluded: failed gates	197040ae95
data/pilot/focused_qwen_v2	54	Q7	S0-S2; C/O/T	v1	excluded: incomplete passing subset	f2b82cc82c
data/pilot/latest	54	Q7	S0-S2; C/O/T	v1	excluded: incomplete passing subset	ce3f846149
data/pilot/llama32_3b_neurips_probe_v1	45	L32	S0-S2; C/O/T	v1	excluded: failed gates	84add58618
data/pilot/llama_calendar_attendee_full_v1	4	L31	S0; C/O/T	v1	excluded: no validity report	357e1815ff
data/pilot/llama_calendar_attendee_v3	18	L31	S1/S2; C/O/T	v1	excluded: failed gates	e8fbbdb08e
data/pilot/llama_neurips_probe_v1	6	L31	S0; C/O/T	v1	excluded: failed gates	7572a3da47
data/pilot/llama_rescue_toolfix_v1	3	L31	S2; C/O/T	v1	excluded: failed gates	21e6c90568
data/pilot/meeting_time_s0_clean	1	Q7	S0; C	v1	excluded: failed gates	6db65c742f
data/pilot/qwen25_3b_neurips_probe_v1	45	Q3	S0-S2; C/O/T	v1	excluded: failed gates	97714dc685
data/pilot/qwen25_corrected_affected_slice	81	Q7	S0-S2; C/O/T	v1	excluded: failed gates	95f91608db
data/pilot/qwen25_corrected_merged	324	Q7	S0-S2; C/O/T	v1	excluded: superseded 324-row candidate	aff1934e11
data/pilot/qwen_calendar_attendee_confirm_v1	4	Q7	S0; C/O/T	v1	excluded: no validity report	ba463a0767
data/pilot/qwen_expense_toolfix_v1	9	Q7	S2; C/O/T	v1	excluded: failed gates	aa6df82234
data/pilot/qwen_gate_repair_v1_approval_limit	27	Q7	S0-S2; C/O/T	v1	excluded: failed gates	254aaae982
data/pilot/qwen_gate_repair_v1_calendar_attendee_s1	9	Q7	S1; C/O/T	v1	excluded: failed gates	53366ae536
data/pilot/qwen_gate_repair_v1_calendar_location_s0	9	Q7	S0; C/O/T	v1	excluded: failed gates	7a37f2bb61
data/pilot/qwen_gate_repair_v1_meeting_time	27	Q7	S0-S2; C/O/T	v1	excluded: failed gates	1f1d335102
data/results_qwen_repaired_v1	324	Q7	S0-S2; C/O/T	v1	excluded: superseded 324-row candidate	6e1efba10e
data/results_qwen_toolfix_v1	324	Q7	S0-S2; C/O/T	v1	excluded: superseded 324-row candidate	ad891adf60
data/results_qwen_v3_full	94	Q7	S0; C/O/T	v1	excluded: no validity report	d5ad3f928d
data/calibration/s1_oracle_retrieved_memory	72	Q7	S1-ORACLE; T	v1	included calibration-only sensitivity packet; excluded from main rates	6d8542da44

346 G Full Related-Work Comparison

Table 13: Detailed positioning of MEMTRACE relative to related benchmarks and attacks.

Benchmark/work	Tool actions	Untrusted retrieval	Persistent memory	Delayed trigger	Separates admission vs executed violation	Execution-failure accounting	Open trace release
ToolEmu	yes	partial	no	no	no	partial	no main trace requirement
AgentDojo	yes	yes	no	no	no	partial	benchmark code/data
PoisonedRAG	no/limited agent tooling	yes	RAG store	retrieval-time	no	no	attack-focused
AgentPoison	yes	yes	yes/RAG memory	yes	no	no	attack-focused
MINJA	yes	yes	yes	yes	no	no	attack-focused
Memory Arena	yes	not security focus	yes	multi-session utility	no	no	benchmark
Memory AgentBench	varies	not security focus	yes	multi-turn utility	no	no	benchmark
MEMTRACE	yes	yes	yes	yes	yes	yes	yes

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [\[N/A\]](#)

Justification: The benchmark does not include crowdsourcing experiments or research with human subjects.

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661 Justification: The benchmark uses synthetic tasks and generated traces, not human-subject
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